Sound: pitch and amplitude

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a longitudinal-wave model to distinguish pitch and sound level.

SUCCESS CRITERIA - I can...

1. Describe compressions and rarefactions.
2. Connect pitch primarily with frequency.
3. Separate particle motion from sound propagation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “pitch” and “amplitude” to compare your model or result. Sentence starters are optional.

pitch: the perceived highness or lowness of a sound

amplitude: maximum displacement from equilibrium for the wave

longitudinal: a disturbance parallel to the wave’s travel direction

01 Bring back an earlier idea

A wave enters a medium where its speed is 3 times as large, at a small angle below critical conditions. Does the transmitted ray bend toward or away from the normal?

Word help: medium = the material through which a mechanical wave travels

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $T = 1/f$; $v = f\lambda$.
Higher f generally means higher pitch. Amplitude affects intensity under matched conditions, not frequency.

Worked reasoning: A tuning fork makes 440 cycles each second, so its frequency is 440 Hz. A louder version can have the same frequency but greater amplitude.

Your turn: Draw two displacement-time traces with equal periods and different amplitudes.

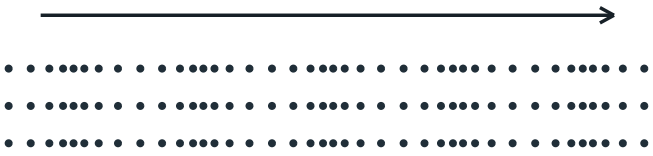
Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

03 Look and explain

Mark compressions and rarefactions in the particle picture. Do air particles travel from speaker to listener with the disturbance?

Sentence starter: I notice … . This supports … because … .



Label dense and sparse regions.

Marzano check - circle: ① ② ③

04 Build the solution

Tone A is 200 Hz; B is 400 Hz. Which has higher pitch? If both play for 2 s, how many cycles does each produce?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Use the idea

Turning up a speaker’s volume does not normally double a song’s pitch.
Explain which wave property mainly changes.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Choose a method

A refrigerator uses 24 J electrical work while moving heat from its colder interior to a warmer room.
Explain why this is not spontaneous cold-to-hot heat flow.

Word help: spontaneous = able to occur without the specified external work input

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

07 Exit check

Is sound in air mainly transverse or longitudinal?
State how particles move relative to propagation.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Design two sound changes: one changes pitch only and one changes amplitude only.
Describe how their graphs differ.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3